

# Production Functions Problem Set

ECON101—INTRODUCTION TO MICROECONOMICS.

FALL 2023

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## Problem 1

For each of the following production functions, please sketch a representative isoquant at level of production  $q_0$ . Also calculate the marginal physical product for each input, and determine whether each marginal product is diminishing, constant, or increasing. Then calculate the marginal rate of technical substitution for each production function. Determine whether the production function exhibits constant, increasing, or decreasing returns to scale.

(a)  $F(L, K) = LK^3$

(b)  $F(L, K) = L + 3K$

(c)  $F(L, K) = (\min\{L, K\})^{\frac{1}{3}}$

## Problem 2

The production function for fragles is  $F(K, L) = L/2 + \sqrt{K}$ , where  $L$  is the amount of labor used and  $K$  is the amount of capital used.

- (a) Determine the returns to scale of the production function.
- (b) Consider the marginal physical product of labor. Is it constant, increasing, or decreasing?
- (c) In the short run, suppose capital is fixed at 4 units. Labor is variable. Determine and graph the short-run production function of fragles as a function of the amount of labor used. Draw it in a diagram.
- (d) Same short-run case as above: Compute the marginal physical product of labor, the average product of labor and draw them in a diagram.

## Problem 3

The firm Muppets Inc., has two plants for producing widgets, one in Sesame Street and another one in Fraggle Rock. The Sesame Street plant produces according to

$$F_S(x_1, x_2) = \min\{x_1, 2x_2\},$$

and the Fraggle Rock plant produces according to

$$F_F(x_1, x_2) = \min\{2x_1, x_2\},$$

where  $x_1$  and  $x_2$  are inputs of production.

- (a) Draw the isoquant for 40 widgets for the Sesame Street plant. In the same diagram, draw the isoquant for 40 widgets at the Fraggles Rock plant.
- (b) Suppose that the firm wishes to produce 20 widgets at each plant. How much of each input will the firm need to produce 20 widgets at the Sesame Street plant? How much of each input will the firm need to produce 20 widgets at the Fraggles Rock plant? Define input combination  $A$  as the total amount of each input that is needed to produce 20 widgets at the Sesame Street plant and 20 widgets at the Fraggles Rock plant. Plot this point in the diagram that you made before.
- (c) Suppose now that the firm wishes to produce 10 widgets at the Sesame Street plant and 30 widgets at the Fraggles Rock plant. Define input combination  $B$  as the total amount of each input that is needed to produce 10 widgets at the Sesame Street Plant and 30 widgets at the Fraggles Rock plant. Plot this point in the diagram you made before.
- (d) Do the same as in the previous two tasks but now supposing that the firm wishes to produce 30 widgets at the Sesame Street plant and 10 widgets at the Fraggles Rock plant. Denote by  $C$  the input combination representing the total inputs needed and plot this point in the diagram you made before.
- (e) Assuming that the firm can pass inputs freely between the two plants, What is the firm's isoquant for producing 40 widgets if the firm can split production in any manner between the two plants?